ETSI GS OEU 005 V1.1.1 (2018-07)

Operational energy Efficiency for Users (OEU); Data Management for Global KPI of ICT Site Energy Management

Disclaimer

The present document has been produced and approved by the Operational energy Efficiency for Users (OEU) ETSI Industry Specification Group (ISG) and represents the views of those members who participated in this ISG.
It does not necessarily represent the views of the entire ETSI membership.

Reference
DGS/OEU-005

Keywords
energy efficiency, user

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - NAF 742 C
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° 7803/88

Important notice

The present document can be downloaded from:
http://www.etsi.org/standards-search

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the only prevailing document is the print of the Portable Document Format (PDF) version kept on a specific network drive within ETSI Secretariat.

Users of the present document should be aware that the document may be subject to revision or change of status.
Information on the current status of this and other ETSI documents is available at
https://portal.etsi.org/TB/ETSIDeliverableStatus.aspx

If you find errors in the present document, please send your comment to one of the following services:
https://portal.etsi.org/People/CommiteeSupportStaff.aspx

Copyright Notification

No part may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm except as authorized by written permission of ETSI.
The content of the PDF version shall not be modified without the written authorization of ETSI.
The copyright and the foregoing restriction extend to reproduction in all media.

© ETSI 2018.
All rights reserved.

DECT™, **PLUGTESTS**™, **UMTS**™ and the ETSI logo are trademarks of ETSI registered for the benefit of its Members.
3GPP™ and **LTE**™ are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners.
oneM2M logo is protected for the benefit of its Members.
GSM® and the GSM logo are trademarks registered and owned by the GSM Association.

Contents

Intellectual Property Rights 4
Foreword 4
Modal verbs terminology 4
Introduction 4
1 Scope 5
2 References 5
2.1 Normative references 5
2.2 Informative references 5
3 Definitions, symbols and abbreviations 6
3.1 Definitions 6
3.2 Symbols 6
3.3 Abbreviations 6
4 Energy management of ICT site 6
4.1 Generalities 6
4.2 Site information 6
4.3 Energy consumption stats 6
4.4 Site reports 7
5 Site information 7
5.1 Generalities 7
5.2 Site information inputs 7
5.2.1 Location 7
5.2.2 Built age 7
5.2.3 Area 7
5.2.4 Tier Class 7
5.2.5 Label 8
5.2.6 Assessment intermediary 8
5.3 Capacity information 8
5.3.1 DC Power capacity 8
5.3.2 Cooling capacity 8
5.3.3 IT power capacity 8
6 Energy management stats 8
6.1 Generalities 8
7 Site reports 9
7.1 Generalities 9
7.2 Mandatory Site Report inputs 10
7.2.1 Year of data record 10
7.2.2 EC_{DC} 10
7.2.3 EC_{SP} 10
7.2.4 EC_{REN} 10
7.2.5 EC_{REUSE} 10
7.2.6 EC_{TH} 10
7.2.7 EC_{HE} 10
7.3 Annual report summary 10
Annex A (informative): Optional report inputs 12
History 13

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables may have been declared to ETSI. The information pertaining to these essential IPRs, if any, is publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI Web server (https://ipr.etsi.org/).

Pursuant to the ETSI IPR Policy, no investigation, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

Foreword

This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) Operational energy Efficiency for Users (OEU).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Introduction

The objective of the present document is to enable the aggregation of study cases and examples of the implementation of ETSI EN 305 200-3-1 [1].

1 Scope

The present document defines a general process to monitor the energy performance of ICT sites in accordance with ETSI EN 305 200-3-1 [1].

The present document:

- specifies required data to register the ICT site (data centre or any other ICT site) within the data aggregation and monitoring system;
- defines the format for data submission to allow the information for a particular ICT site to be aggregated and monitored;
- describes a tool to calculate the Global KPI for Energy Management (KPI_{DCEM}) as specified in ETSI EN 305 200-3-1 [1].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI EN 305 200-3-1: "Access, Terminals, Transmission and Multiplexing (ATTM); Energy management; Operational infrastructures; Global KPIs; Part 3: ICT Sites; Sub-part 1: DCEM".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

Not applicable

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in ETSI EN 305 200-3-1 [1] and the following apply:

year of data record: year of a data collection period started

3.2 Symbols

For the purposes of the present document, the symbols given in ETSI EN 305 200-3-1 [1] and the following apply:

EC_{DC}	Energy Consumption DataCentre
EC_{FEN}	Energy Consumption Fossil Energy
EC_{REN}	Energy Consumption Renewable
EC_{REUSE}	Energy Consumption Reused
EC_{SP}	Energy Consumption Specific Provider

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

DC	Data Centre
EER	Energy Efficiency Ratio
ICT	Information Communication Technologies
IT	Information Technology
KPI	Key Performance Indicator

4 Energy management of ICT site

4.1 Generalities

Energy management of an ICT site (data centre or any other ICT site) is done in a global screen.

This global screen of ICT site energy management is subdivided in three sub-parts:

- Site information.
- Energy consumption stats.
- Site reports.

4.2 Site information

This sub-part "Site information" describes the main technical characteristics of the site.

4.3 Energy consumption stats

This sub-part "Energy consumption stats" provides global energy consumption management statistics of the site.

4.4 Site reports

This sub-part "Site reports" provides a general summary of yearly reports of the site.

Recorded data are input in the "new report" screen.

5 Site information

5.1 Generalities

Registration of the site is done by description of the ICT site (data centre or any other ICT site) and its main characteristics.

Recorded data are input in the first sub-part of the global screen.

The date spaces of this first sub-part are used to input characteristic data, see figure 1.

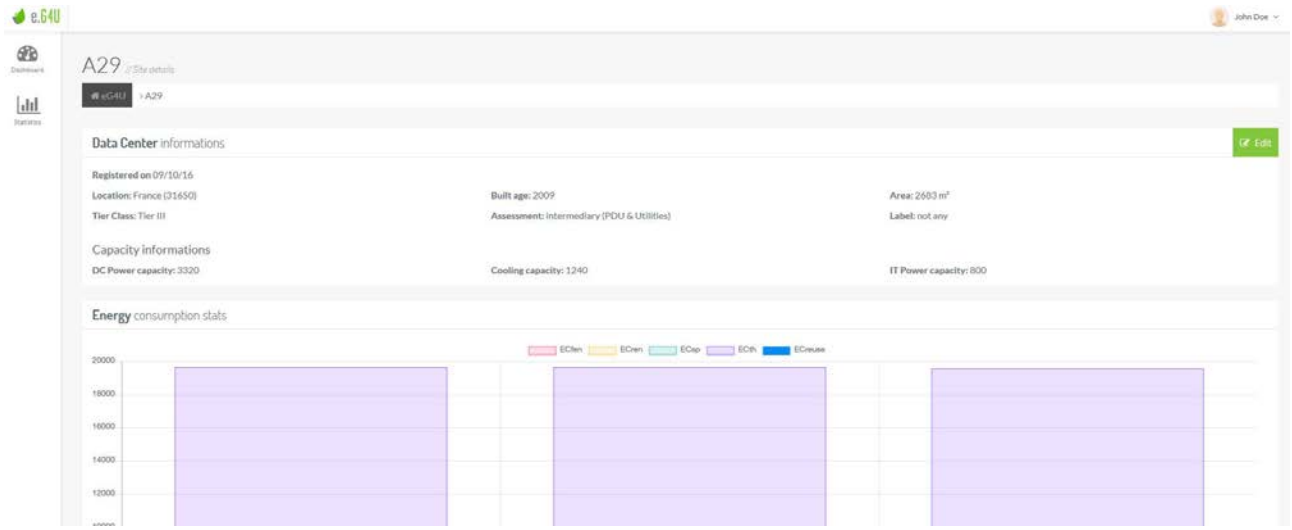

Figure 1: General site information

5.2 Site information inputs

5.2.1 Location

This data space will define the location of the site.

5.2.2 Built age

This data space will define the year of site building.

5.2.3 Area

This data space will define the surface of the site.

5.2.4 Tier Class

This data space will define the Tier class of the site.

5.2.5 Label

This data space defines the site label if there is one.

5.2.6 Assessment intermediary

This data space will define the general use of the site.

5.3 Capacity information

5.3.1 DC Power capacity

This data space defines power capacity of the site.

5.3.2 Cooling capacity

This data space defines cooling capacity of the site.

5.3.3 IT power capacity

This data space defines power capacity of IT equipment.

6 Energy management stats

6.1 Generalities

Energy management statistics are automatically provided in the second sub-part of the global screen.

This second sub-part provides yearly statistics as a global curve of reused energy, see figure 2.

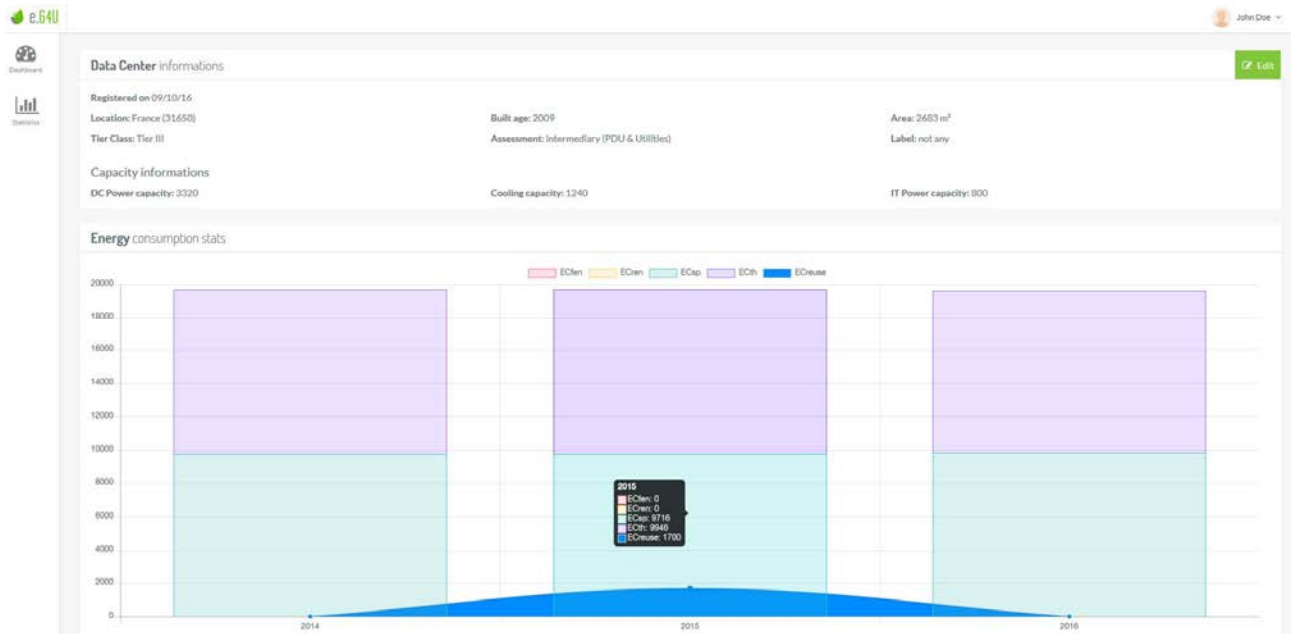

Figure 2: Energy management statistics

7 Site reports

7.1 Generalities

A report is made on the energetic data of the site every year. The annual data are input by opening another screen by clicking on the "new report" button.

Recorded data are input in the "new report" screen.

The date spaces of the "new report" screen are used to input annual data, see figure 3.

In the third sub-part of the main screen a table summarized the site yearly reports, see clause 7.3.

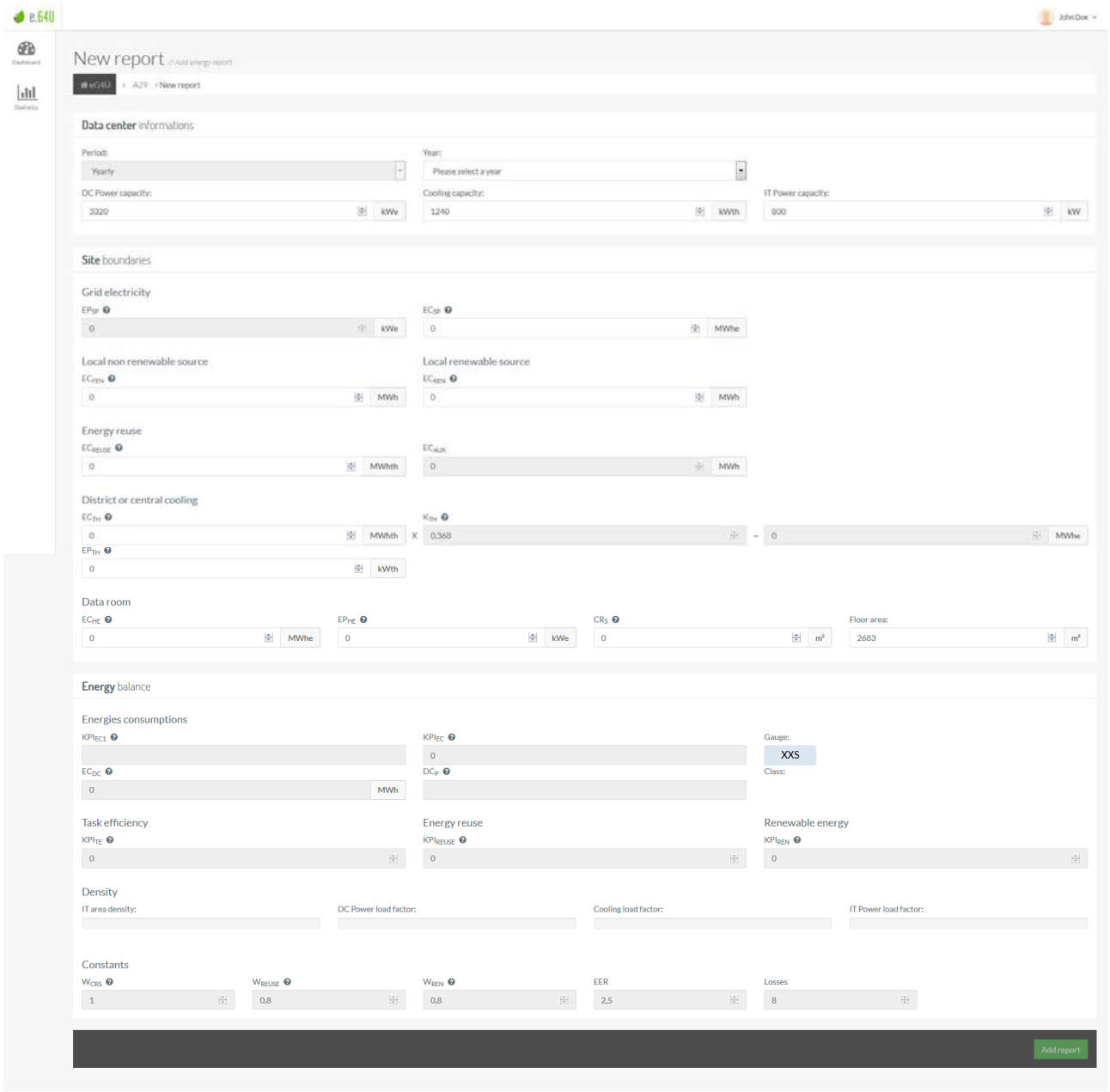

Figure 3

7.2 Mandatory Site Report inputs

7.2.1 Year of data record

This data space defines the year of this data collection period started.

7.2.2 EC_{DC}

This data space defines the Total consumption by Data Centre during the collection period.

7.2.3 EC_{SP}

This data space defines the Total energy consumption from grid electricity during the collection period.

7.2.4 EC_{REN}

This data space defines the Total energy consumed during the collection period from local renewable energy sources.

7.2.5 EC_{REUSE}

This data space defines the Total energy reused during the collection period at the delivery point.

7.2.6 EC_{TH}

This data space defines the Total energy consumed during the collection period from external cold loops.

7.2.7 EC_{HE}

This data space defines the Total energy consumed during the collection period by ICT equipment.

7.3 Annual report summary

In the third sub-part of the main screen a table summarized the site yearly reports as defined in figure 4.

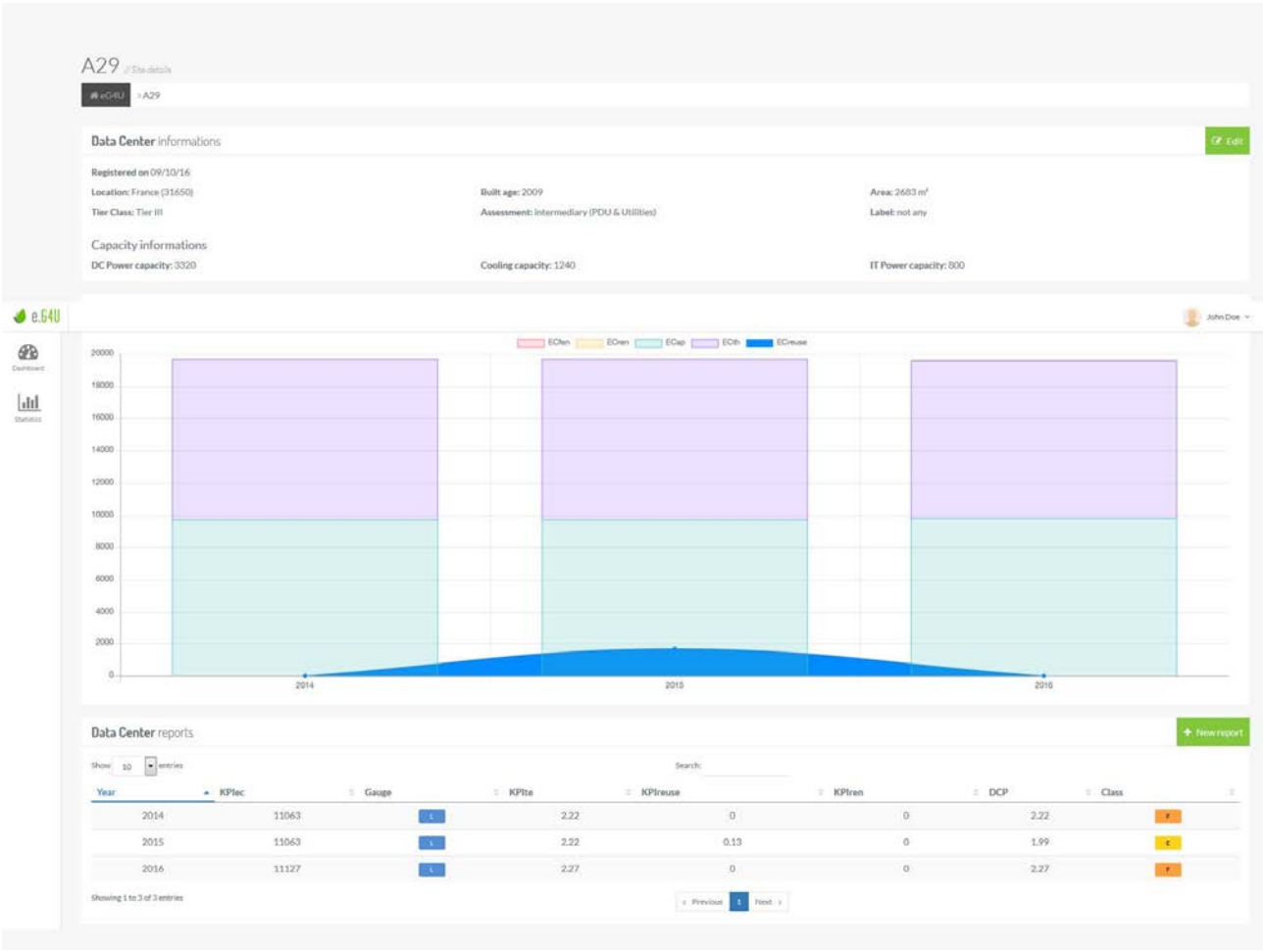
A29
Site details
e.G4U
A29
Data Center informations
Edit
Registered on 09/10/16
Location: France (31650)
Built age: 2009
Area: 2683 m²
Tier Class: Tier III
Assessment: intermediary (PDU & Utilities)
Label: not any
Capacity informations
DC Power capacity: 3320
Cooling capacity: 1240
IT Power capacity: 800
e.G4U
John Doe
Dashboard
Statistics
ECfen
ECren
ECap
ECth
ECreuse
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
2014
2015
2016
Data Center reports
New report
Show 10 entries
Search:
Year
KPIec
Gauge
KPIte
KPIreuse
KPIren
DCP
Class
2014 11063 L 2.22 0 0 2.22 F
2015 11063 L 2.22 0.13 0 1.99 E
2016 11127 L 2.27 0 0 2.27 F
Showing 1 to 3 of 3 entries
Previous
1
Next

Figure 4

Annex A (informative): Optional report inputs

EC_{FEN}

This data space defines consumption of local fossil energy during collection period.

EER

This data space defines Energy Efficiency Ratio of external cold loop cold generator.

Losses

This data space defines losses in the external cold distribution loop.

Density

This data space will define the percentage of IT room occupation.

History

Document history		
V1.1.1	July 2018	Publication